Simulation of demurrage in Threadneedle

by

Birgir Steinarsson

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Abstract

Demurrage, in terms of currency, adds a cost of carry to money which discourages hoarding of money and increases the velocity of money in the economy. We simulate demurrage in Threadneedle as a new tax levied on balances of deposit accounts. The simulation is inspired by the Wörgl Experiment, that enabled a small town to flourish in the Great Depression after issuing a new local currency using demurrage.

Keywords: Demurrage, Computer simulation, Agent based modeling, Economics

Hermun á gjaldmiðilsrýrnun í Threadneedle

Birgir Steinarsson

janúar 2026

Útdráttur

Gjaldmiðilsrýrnun bætir geymslukostnaði við peninga sem eykur veltuhraðann í hagkerfinu og virkar letjandi fyrir uppsöfnun peningamagns. Við útfærum hermun á rýrnuninni í Threadneedle með nýjum skatti sem leggst á innistæður bankareikninga. Hermunin sækir innblástur sinn í Wörgl tilraunina sem leyfði bænum að blómstra í kreppunni miklu eftir að hafa stofnað staðbundinn gjaldmiðil sem rýrnaði með tímanum.

Efnisorð: Gjaldmiðilsrýrnun, Hermilíkön, Einingarhermun, Hagfræði

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Birgir Steinarsson
Master of Science

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Chapter 1

Introduction

Demurrage has been proposed as a fee imposed on money holders, a cost or charge for owning money. The 20th century economist Silvio Gesell, in his book *The Natural Economic Order*[1], proposed a monetary reform using demurrage as a means to discourage hoarding of money and increasing the velocity of money through the economy. He suggested that money owners had an advantage and could withhold spending money which put pressure on producers of perishable goods to sell at a lower price since the value of their goods would decrease faster than the money. In Gesell's view this function of money as a store of value interfered with the function of money as a token of exchange and proposed demurrage as a tool to balance the scales between these two functions.

There have been several real world experiments in implementing demurrage in a monetary system, the most famous of which was in Wörgl, Austria, during the Great Depression. While European banks had severe liquidity problems, the people of Wörgl could not conduct business or trade due to a lack of money. The local municipal government of Wörgl, led by the mayor Unterguggenberger, issued their own local currency in the form of a stamp scrip. The council paid its workers salaries in this new form of money which used the properties proposed by Gesell. Every month people had to get their money stamped and had to pay 1% of the face value as a form of demurrage.

The experiment was considered a success, alleviating extreme unemployment and 200 other townships considered replicating it before it was shut down by the government.[2] It is not clear if the reported success was due to demurrage, or simply because they created a new local currency, but the town of Wörgl was seen as having found a solution to the depression, at least until the experiment was shut down by the National Bank of Austria. In fact, Godschalk suggests that the rate of demurrage was not correlated to the success of the currency where it has been implemented and the addition of the local currency might have sufficed.[3]

In order to explore these issues we implemented demurrage in a simulated economy in Threadneedle, an economic simulation framework based on double-entry bookkeeping in a multi-agent environment, and examined the effects it has on the economy in this simple market based economic simulation. We started by building a baseline economy to compare and contrast with demurrage and tried to cover the many aspects and problems inherent with building a simulated economy.

1.1 Background

Unfortunately, there is not a great wealth of literature covering demurrage in an economic sense as other monetary frameworks have prevailed in mainstream economics. We are not aware of any other attempts to explore demurrage in economic computer simulations except for an agent-based simulation[4] based on the Encointer project, which proposes using a local cryptocurrency with both Universal Basic Income (UBI) and demurrage to empower economically deprived areas with a complementary community currency.[5] Our simulation in Threadneedle differs from this one since Threadneedle does not rely on any mathematical models from economics as opposed to many other economic simulations.[6]

1.1.1 Threadneedle

Threadneedle was created at Reykjavik University as a tool to study the economy and the fractional reserve banking system from the viewpoint of computer science, building the economy as a complex system of network nodes. [7] This allows us to use known aspects of computer networks to analyse a simulated economy, granting a different viewpoint from economic theory. It is a multi-agent framework and a digital twin that attempts to exactly reproduce banking system bookkeeping and accounting operations in order to explore its systemic behaviours, including support for simplified versions of Basel 2 and gold standard (fixed asset) regulatory frameworks. The codebase is written in Java and uses JavaFX as an UI layer to simplify the building of the simulations. It also has a command-line interface that can be used instead and offers greater computational efficiency than the graphical version. Threadneedle is designed to support most regulatory frameworks, such as the Basel accords, to simulate the price determination mechanism of the market-maker markets and analyze the flow of money through the system.

Every agent in the system, whether it is a company or person, government or bank, has a bank account which is used to keep track of income, expenditure, and their running balance. The simulation can only continue to run as long as the flow requirements are met. That means that all agents must have an income stream to be able to sustain their output stream. As we will see later on, in a constant money economy, the difference between the input and output streams must, in the long run, converge to zero for a simulation to keep running, since it will essentially be a zero-sum game.

1.1.2 Demurrage

Demurrage is a term widely used in the shipping industry as a term for fees related to delays and detentions, e.g. if a ship stays in a port for a period of time that is longer than a contract allows, although according to Storms et al. it doesn't have a precise definition, even within the industry.[8] This definition of demurrage is not our focus but there are some similarities to the currency related meaning. Currency demurrage is defined as a time and/or transaction fee based on the nominal value of a currency, paid by the holder of the currency, and can therefore be viewed as a fee for the delay of money. Depending on how the demurrage is implemented, it can be a depreciation where the nominal value decreases over time and therefore decreases the total money supply or in the form of taxation where the money supply is constant but the fee is

paid to the government. In the case of depreciation, if the money supply expands at the same rate as the depreciation, whether that is due to government printing, bank lending or crypto-mining, then the total money supply would also be constant and effectively be an equivalent implementation.

There have been several implementations of currencies that include demurrage, many of which had been introduced as emergency money in the form of stamp scrip during a crisis, both in Europe and America. Depression-scrip was common in the U.S. but mostly used transaction based demurrage, where the value would decrease every time the money changed hands. Irving Fisher published *Stamp Scrip* in which he tried to convince people to use time based demurrage instead, because that would increase the velocity of the currency which fit into Fisher's economic theories. Fisher advocated for this in Congress but was unsuccessful, yet some U.S. communities tried issuing time based scrip.[9] The main difference between the methodology between Fisher and Gesell was that Fisher only viewed this currency as a temporary solution to be used in a time of monetary crisis, where Gesell proposed it as a general feature of money. This resulted in Fisher suggesting a 2% weekly fee that would devalue the currency totally in a single year, but Gesell intended a 5% annual rate, only to counteract the inherent value of money due to its liquidity advantages. All of the demurrage currencies that have been attempted have been implemented as a complementary currency but never as the main legal tender of a country.[3]

Several methods of implementing demurrage have been suggested by Gesell and others but the technology of the time posed some limitations. *Table-money* was one suggestion where a bill would have printed on it dates which would show decreasing value as time passed, for instance it could have 52 values, one per week of the year, where each value would lower the original nominal value by 0.1%, which would amount to a total loss of 5.2% after the year. After that the notes would no longer be valid and would have to be issued again. Giving back change was also a problem for paper money and *Series-money* was a suggested solution where smaller denominations could be issued in different colored series and each year a randomly selected color would be deemed expired and lose all value. Presumably, 20 different color series would have to be produced so that one color being devalued would mean a 5% demurrage rate on average.[1]

The most popular solution, which was used in Wörgl, was *Stamped-money*, where people would have to buy stamps to affix to the bills periodically for the bill to retain its nominal value. The cost of the stamp would be the demurrage percentage rate and could be bought within the same currency but often the state currency was used along with or instead for amounts that didn't exist in denominations of the demurrage currency.

Current technology provides much more efficient alternatives, at least with digital versions like bank deposits and other digital only currencies like cryptocurrency. Demurrage fees could be collected automatically from bank accounts periodically, whether that would be monthly, daily or even shorter periods. Currently, banks sometimes show depositors two different numbers for their account balance, one being the registered balance and another "redeemable" balance that includes transactions that are still pending in the banking system. This might be used in a similar way to show the

[1] Perhaps some clever arrangement of sub-denominations, e.g. 10, 5, 1, cents or pfennigs, could be assigned to fewer colors and still provide a 5% demurrage but we will leave that as an exercise for the reader.

balance after the last transaction but the redeemable amount would be less due to the amount that has been devalued due to demurrage, instead of showing periodical transactions each time the demurrage is applied. With this implementation even a continuous rate of demurrage would be feasible but since other interest calculations, e.g. savings or loan interests, are not being calculated in this way it is unlikely that demurrage would be. Another idea would be to borrow an accounting FIFO process from inventory valuation methods, where each transaction from an account would retroactively calculate the demurrage fees due on the account, depending on how long each "quant" of money has been sitting in the account and remove the oldest ones first. Any implementation might be susceptible to some exploitation, e.g. if the intention of demurrage is to impose a fee for the delay of money, employers might pay salaries the day before demurrage collection so the holding fees get paid by the employees instead, even if they have not been holding the money.

In our simulations in Threadneedle we will not have to worry about these exploits as our agents don't have the ability to utilize them. Instead we simply collect a percentage of all deposits in each step of the simulation.

1.1.3 Experiments with demurrage

We found some conflicting[10] sources[3] regarding where the first experiment with Gesell's demurrage currency[2] was attempted but they all point to Germanic regions around the time of the Great Depression.

One of the earliest, if not the first, was the Wara currency in Schwanenkirchen, Bavaria, Germany, in 1931. The town was mostly based around a single coal mine and when it shut down during the GD the owner issued his own currency to pay the workers. The demurrage rate was 1% per month and was backed by coal from the company. The experiment lasted two years before German authorities shut it down citing risk of hyperinflation.[10]

In America, many stamp scrip currencies were issued but most of them were devalued per transaction, some had both time and transaction based demurrage but only around 5% of them were just time based. The success of the currencies varied but they were designed as a temporary stop-gap measure during crisis.

All of these experiments had something in common, they were too short-lived, either due to government opposition or by design, to be able to draw any conclusion about how demurrage affects the economy in the long run. They were also all issued as complimentary currency in a depression where people had little choice other than to accept this new currency.

There are some more recent examples of demurrage based currencies, Chiemgauer started in Bavaria in 2003 as part of the Regiogeld network[12, p. 318] and is still active. It is a stamped money implementation with a 2% demurrage fee every 3 months and is used by local companies and non-profit organizations, where the non-profit get better exchange rates than businesses.

A number of cryptocurrencies also exist that implement demurrage. Freicoin is a cryptocurrency founded in 2011 which has both a transaction and time based demurrage[13]. Circles was launched in 2020 with demurrage and Universal Basic Income but we have not found any sources on how successful it has been.[14]

[2]There are was also the bracteate system in the middle ages where coins were intermittently recalled and recoined in lesser amount, where the difference was seigniorage for the minting authorities.[11]

Figure 1.1: Wörgl Labour Certificates

1.1.4 The Wörgl experiment

The most well known example of currency demurrage being used in a real world scenario comes from the small Tyrolese town of Wörgl in Austria, population around 4000 in the year 1929. During the Great Depression, the municipal government decided to create a new currency, based on the works of Silvio Gesell, in an attempt to alleviate the hardship that the townsfolk were suffering and restart production in the economy. [15] The municipality had debts it could not pay since it was not receiving tax payments. At auctions people who had money decided they would rather keep the money, betting on prices going down further. Withholding money apparently had greater value to people than spending it in these circumstances which indicates a quantitative monetary deflation was in progress.

The municipality established a welfare committee which would issue Labour Certificates whose monetary value was pegged to the Austrian Schilling. The Wörgl government would pay workers with these certificates, mostly for communal work like building bridges, fixing roads etc. The workers in turn could pay their taxes using this local currency in lieu of the national Schilling. Additionally, every month the holders of these certificates would have to purchase stamps from the government at a 1% rate of the denomination of their certificates, lest they would suffer a loss from their face value. It is unclear from the sources we have found from this period whether this fee would have to be paid in a specific currency or if one could pay the demurrage fee of the certificates using the certificates themselves.

A total of 32.000 in nominal value of bills were printed by the welfare committee but the maximum that reached circulation was 7443. [15] Skeptics of the program pointed out that it must be a hoax since even if only 1.000 Schillings in these bills had been issued, a total of 5.100 in tax debts had been paid in a few days. This implies that the circulation of bills was clearly substantial, since for every received tax payment, the municipality was able to pay other invoices and so on.

The inscription on the bills in English translation is the following:

> TO ALL,
>
> Slow circulating money has thrown the world into a tremendous economic crisis and millions of hard-working people into dreadful suffering. The demise of the world from pure economic viewpoint shows its terrible beginning. It is time to save the downward rolling economic machine with sound recognition and determined action, so that mankind isn't driven into brother-wars, confusion and disintegration. The people live by exchanging their goods and services. The slow circulation has severed the exchange of goods and services to a great deal and thereby millions of citizens, willing to work, have already lost their sphere of living in the economic realm. Therefore the exchange of goods and services has to be increased again and the sphere of living regained for those who have been already deprived of it. To this end the certificates of work done notes of the small town of Wörgl are issued:
>
> IT ALLEVIATES THE MISERY AND GIVES WORK AND BREAD.

Chapter 2

Methods

To ascertain the effects of implementing demurrage into an economy we first need to establish a baseline to compare it with. In Threadneedle there are multiple variables that we need to configure to set up our economy. Every simulation in Threadneedle contains at least one instance of the following agents: Government, Central Bank and a Commercial Bank. We will only use the commercial bank to provide other agents with a deposit account to hold their money and transfer between them. The Central bank will be used to provide the Government agent with a bank account but in other aspects it will function the same as the Commercial Bank. The Government agent contains many of the configurable variables we will use during the experiment, e.g. Personal and Corporate Income Tax Rates and the Demurrage Rate we implemented for this project. The banks will not be creating loans in this simulation, i.e. we have a non-lending (or constant money) economy where the money supply is fixed for the duration.

Demurrage was implemented in a similar way to income tax except that instead of taxing income for each period, it taxes the balance of the deposit accounts of the agents. The rate is the same for personal and corporate accounts.

2.1 Building the baseline

Designing the baseline economy for the simulation turned out to be the most challenging part of this thesis. There are multiple variables that need to considered and decided for every Threadneedle simulation, which variables to include, choosing the value of each included variable and which variables should be adjusted between simulations to compare and contrast the results. For computer scientists with no real background in economic theory it is easy to underestimate the complexity of the economic systems of the world. Creating a faithful simulation of the Wörgl economy in Threadneedle to compare with and without demurrage is not possible and maybe not even be desirable since it would make it harder to interpret the results. Threadneedle allows us to build simplified simulations from first principles to explore economic mechanisms like market behavior and money flow.

To be able to build our baseline economy we first needed to understand how the simulations work and decide which variables we consider relevant so we can interpret the results. In the following subsections we explore the variables and our reasoning for deciding on how to build the baseline.

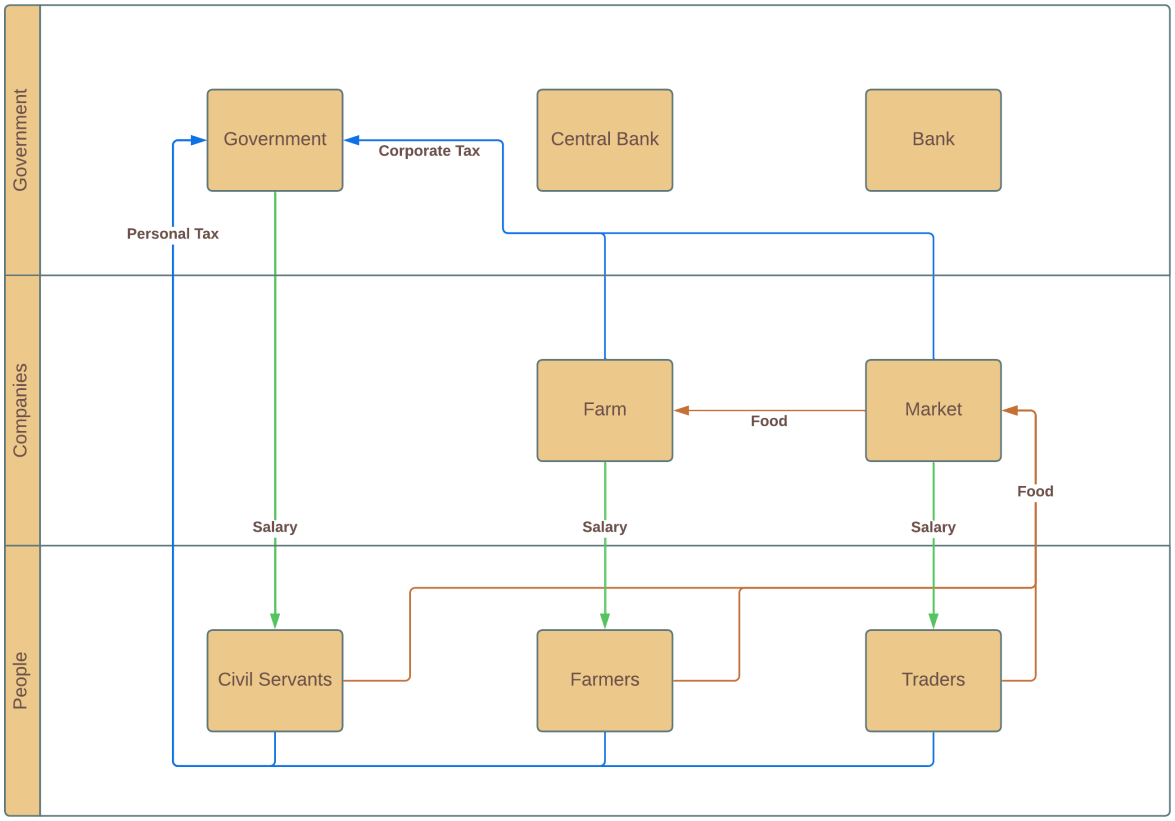

Figure 2.1: Network diagram

2.1.1 Constant Money Economy

Contrary to a lending economy, where money is created as deposits by bank lending and removed by loan payments and loan defaults, our simulation will have a fixed amount of money during the simulation. Every transaction from one account to another has a net of zero in the simulation as a whole. Therefore it follows that if any one agent is accumulating money over time, that money is not available to other agents. For an economy, using a monetary system, to function it needs to provide a way for agents to both receive and spend money to allow for continuous flow for money throughout the system. If we expect all of our agents to be able to continuously participate in the system, i.e. buy food, pay salaries and not go bankrupt, the market mechanism must be able to find a solution where no one accumulates too much money in the long run.

In the simplest of economies it is trivial to calculate a price solution by hand so that this balance is achieved. For larger systems this method quickly becomes unfeasible but as a thought experiment to help us understand the flow problem we work out the calculation for a tiny economy.

Let's imagine that we have a Farmer that works at the Farm and a Trader that works at the Market. These are 4 agents, of which 2 are companies and 2 are people. The Farmer can produce 2 units of food per day, enough for both of them. The Farmer produces the food for the Farm which pays them a daily salary. The Farm then sells the food to the Market which in turn sells the food to the people. The Market also pays the Trader their daily salary. Both the Farmer and the Trader must buy the food from the Market to consume. Then the question becomes, at which prices does the Market buy and sell the food?

Agent	Input	Output
Farmer	X	S
Farm	$2 \times B$	X
Market	$2 \times S$	$2 \times B + Y$
Trader	Y	S

Table 2.1: Trivial Economy

Let's look at a daily cycle in this economy, where B is the price the Market buys each unit of food and sells each unit at S price:

1. The Farmer produces 2 food for the Farm which pays them their X salary
2. The Market buys the 2 food from the Farm for $2 \times B$
3. The Market sells the 2 food to the Farmer and Trader for S each, total of $2 \times S$
4. The Market pays the Trader their Y salary

Since we know that this cycle will repeat endlessly and if any of the agents receive more money in a single cycle than they spend, e.g. the market increases the sales price or the farm lowers the farmers' salary, they will accumulate the money until the simulated economy ultimately fails due to liquidity issues.

Therefore for a continuous flow solution each agent's input must be equal to their output. The Farm will then have to pay the full sale price of the food to the Farmer, ergo $X = 2 \times B$. The Market has to pay the full difference between the sale proceeds and purchasing costs as salary, so $Y = 2 \times S - 2 \times B$ but the Trader has to afford to buy the food (and indeed exactly so) and that is why $Y = S$. Now we have $S = 2 \times S - 2 \times B$ which finally means that $X = Y = S = 2 \times B$.

In this tiny economy it would absolutely be feasible to have a centrally planned economy to decide the prices but as the economy grows larger so does the network information problem which makes a central planning strategy unfeasible. We can also see that the price amount makes no difference as long as the sale price is double the purchase price. In a real-world scenario the multitude of variables quickly obfuscate the problem to a degree where central planning becomes a lot less viable strategy. We also usually don't have a constant money system in the real world, but a (mostly) ever expanding money supply from continuously increasing lending.

We have made the assumption here that the Market will buy and sell each unit of food at the same price within the cycle, which isn't strictly necessary. But even if we remove that assumption the fact still remains that the total sales profit of the Market must exactly match the total salary cost of the Market's employees, in this trivially sized economy.

We already know that we have a constant amount in our money supply but the next problem is deciding on the actual amount. Since we will be using percentage amounts of the monetary values, both transfers and deposits, we might run into some rounding issues. If the personal income tax is set to 20% and a worker's salary for a given period is below 5 units, the worker will not pay any tax on that income. This becomes a bigger problem if we want to set the demurrage rate to 5% or even lower, maybe down to 0.5% per period, but that could be solved with a larger money supply. Also, the Market's spread rate is given in absolute difference between the bid and sell price and not as a ratio, as companies often do for markup calculations. So to scale

up our economy, in addition to adding more agents, we will increase the quantities of food produced and consumed by at least an order of magnitude. This will allow us to work with larger amounts with non-negligible percentages.

2.1.2 Distribution of Labour

In our trivial economy we chose to have the supply of food exactly match the demand. This is not a natural assumption but we will still have to specify the labour output of the farms, which will represent the production of all basic necessities of our population. Each person agent in Threadneedle has a consumption variable to specify how much food they consume per time period. The total value of this variable over every person in the simulation will be the total demand for food in the economy. Therefore, whatever value we choose as the labour output of the farm per employee, will control the ratio needed of the total workforce to satisfy the total food demand. In our previous example the Farmer and the Trader both consumed 1 unit of food per period so we chose the labour output as 2. In that case the workforce ratio required to supply the population with food was 50%. In general, assuming uniform values, the ratio will be equal to the consumption over labour output, e.g. if each person must consume 10 units per day and each person can produce 50 units per day, 20% of the population must be farmers to supply the demand.

If we are satisfied with this abstraction, ignoring real-world concerns regarding the definition of necessity, we will still have to find some work for the remaining population, so as to provide them with input and output flows of money. In the case of the Wörgl experiment, most of the labour purchased by the local government went towards communal infrastructure. We can represent this in Threadneedle with civil servants who don't have a specific labour output to be consumed but are still regarded as providing the economy with some value. We can choose the maximum percentage of the population to be allowed as civil servants and we can already see that we might want that to be all the people that are not farmers or traders. But we can also choose not to limit the ratio and allow the government to employ up to 100% percent of the population.

2.1.3 Time abstraction

Threadneedle runs on time periods that can be specified to simulate real-world periods where employees get a monthly paycheck and have yearly interest rates. For simplification we will process all of the steps in each time period, which means we could interpret it as everyone having a daily paycheck or the amounts produced and consumed are the total amounts during each month. In relation to income taxes this will not affect the calculations, since e.g. 20% of a daily paycheck of 10 money is the same as 20% of a 300 money monthly paycheck. It doesn't matter how many time periods we look at, the percentage yield the same result. However, we have to be careful when interpreting the percentage rate of the demurrage since it will compound over the periods. For example when Silvio Gesell suggested a 5% demurrage rate it is understood to be an annual rate. If we set the demurrage rate as 5% per time period in Threadneedle then it matters if we think of each period as a day, month or a year. A 5 percent monthly rate will compound into an 46 percent annual rate.

2.1.4 Working simulation

We have now covered most of the decisions we face when building this baseline simulation in Threadneedle. But we still haven't defined what constitutes a working simulation. Since we are approaching this economic simulation as a complex networking system we consider simulations that can maintain a continuous flow longer as more successful than those that stop flowing due to accumulation or other factors.

2.2 Implementing demurrage

There are many ways to implement demurrage, such as the stamped money that was used in Wörgl, but we choose to create a new tax directly on deposits. Every deposit account in the banking system will be subjected to demurrage in the form of a percentage tax on the deposited amount. The agents in Threadneedle are not programmed to be able to exploit the mechanism of this system, whereas in the real world more precautions would need to be taken. It might be beneficial for some parties to delay certain transactions until after the demurrage has been collected, for instance. A possible solution to this would be to use the average amount deposited in the period between collections. A similar mechanism to how banks calculate interest payments on deposits could probably be used.

Contrary to the stamped money, where the demurrage does not compound per bill but is always the same amount, our implementation will compound. A stamped bill with the nominal value of 100 and a demurrage fee of 5% per month is worth exactly 40 after one year. A deposit account with the balance of 100 and same demurrage rate will have 54.04 left at the end of the year.

Since the demurrage will compound over time, we have to calculate the nominal rate, per time period t, from our desired compounded rate. We observe that if we have a balance of x_0 and nominal demurrage rate r_n that the deposit will decrease at the following rate:

$$\begin{aligned} x_1 &= x_0 \cdot (1 - r_n) \\ x_2 &= x_1 \cdot (1 - r_n) = x_0 \cdot (1 - r_n)^2 \\ &\vdots \\ x_t &= x_{t-1} \cdot (1 - r_n) = x_0 \cdot (1 - r_n)^t \end{aligned} \tag{2.1}$$

To find the compounded rate we find the percentage that we have paid over these t periods:

$$r_c = \frac{x_0 - x_0 \cdot (1 - r_n)^t}{x_0} = 1 - (1 - r_n)^n \tag{2.2}$$

And since we usually want to refer to the compounded rate as the demurrage rate we isolate the nominal rate as a function of the compounded rate:

$$r_n = 1 - (1 - r_c)^{\frac{1}{n}} \tag{2.3}$$

Imposing an annual compounded rate of 5% collected daily, the nominal rate becomes 0.014% (assuming 360 bank days in a year). Since this would cause problems

Government		Market	
Personal Income Tax (%)	Tax_p	Max. Employees	Emp
Corporate Income Tax (%)	Tax_c	Max. Inventory	Inv
Minimum Salary	Sal_{min}	Price Spread (min:max)	Spr
Max. Civil Servants (%)	Civ	Initial Price (ask:bid)	Pri
Demurrage Rate (%)	Dem		
Farm		**Person x10**	
Labour Output	Lab	Initial Salary	Sal
		Food Purchased	Q_p
		Food Consumed	Q_c
		Food Stored	Q_s

Table 2.2: Simulation Variables

Government		Market		Farm		Person x2	
Dep	0	Dep	30	Dep	0	Dep	0
Tax_p	0	Emp	1	Lab	2	Sal	2
Tax_c	0	Inv	3			Q_p	1
Sal_{min}	0	Spr	1:2			Q_c	1
Civ	0	Pri	4:5			Q_s	1
Dem	0						

Table 2.3: Trivial Simulation

with rounding amounts, the minimum balance would have to be over 7000 to be subject to the fee. Instead we can simply decide that each step in the simulation is one month and for convenience choose the annual compounded rate of 6% for a nominal rate of 0.5% per month. This value can then be adjusted for different experiments. In every step of the simulation each account holder will pay the demurrage rate of their current deposit in the bank.

2.3 Simulations

Threadneedle contains many variables regarding the banking system and fractional reserve, such as interest rates, reserve requirements, loss provisions, that we will be ignoring for our simulations. The variables we will be working with, arranged by the agent they belong to, are listed in table 2.2, along with abbreviations. Additionally, all agents will have an Initial Deposit (Dep), the sum of which will constitute the total Money Supply (M_0).

2.3.1 Simulation: Trivial economy

The initial variables for this simulation in Threadneedle are shown in table 2.3 and the full json file is seen in listing A.2.

In figure 2.2 the market finds the solution that allows for constant production and consumption. [1] The final bid:sell prices end up being 1:2, which matches our previous

[1] The Market keeps firing and hiring the Trader but that has to do with the heuristics it uses to calculate if it can afford to keep its employees.

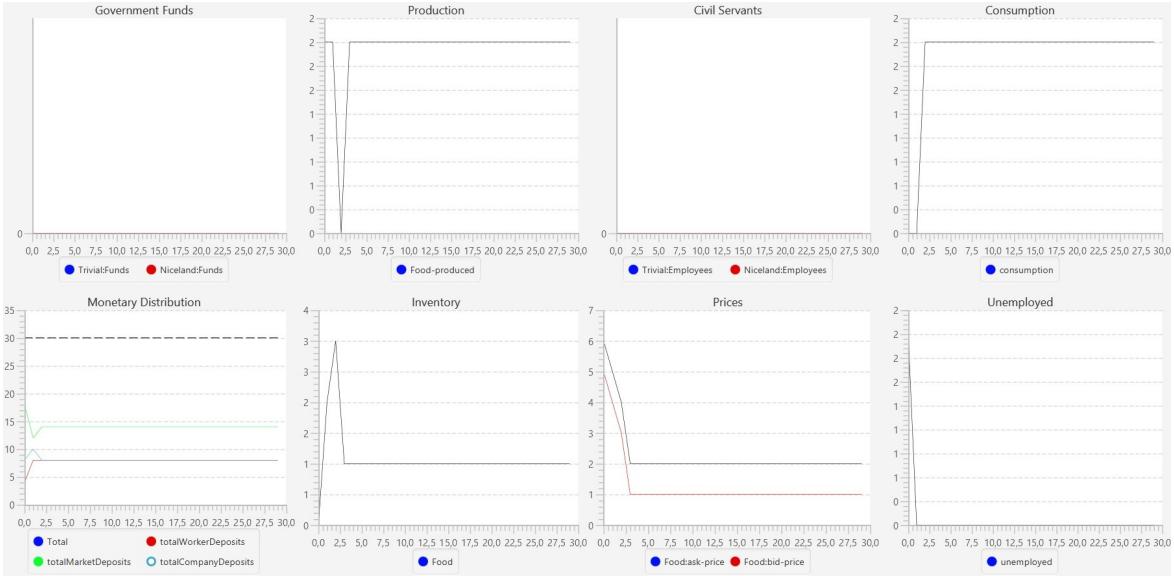

Figure 2.2: Trivial Economy

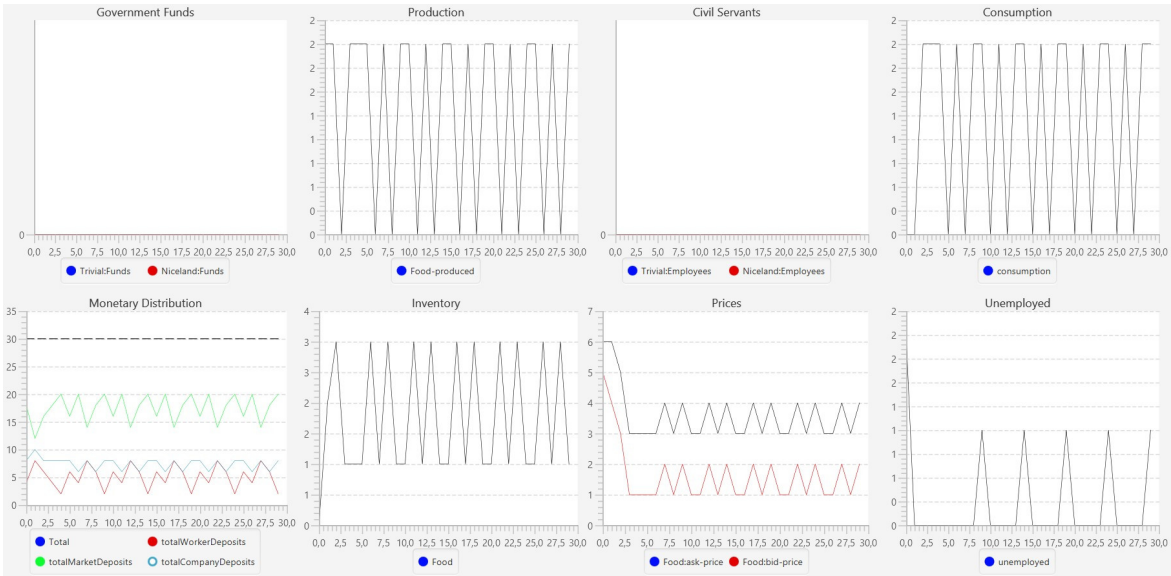

Figure 2.3: Trivial Economy with max spread 3

calculations. We do have to grant some leniency in the beginning of the simulation, while the workers are unable to afford the food, for the market to react and lower the prices. After that, equilibrium is reached and the balance of each account is the same at the end of each step.

Almost any deviation from these initial variables will cause the solution not to be found. We can increase the initial deposits of the agents but that will just increase the time before equilibrium is reached. Most notably, if we allow the max spread to be higher than 2, e.g. 3 the simulation fails, as we can see in figure 2.3. There is no longer constant consumption because the workers run out of funds and cannot afford the food.

It's interesting to see what happens if we scale up the supply and demand by a factor of 10. In table 2.4 we have multiplied almost all numbers by 10 but if we run that simulation it quickly fails, see figure 2.4. This constitutes a partial solution

Government		Market		Farm		Person x2	
Dep	0	Dep	300	Dep	0	Dep	0
T_p	0	Emp	1	Lab	20	Sal	20
T_c	0	Inv	30			Q_p	10
Sal_{min}	0	Spr	1:2			Q_c	10
Civ	0	Pri	4:5			Q_s	10
Dem	0						

Table 2.4: Trivial Simulation 10x

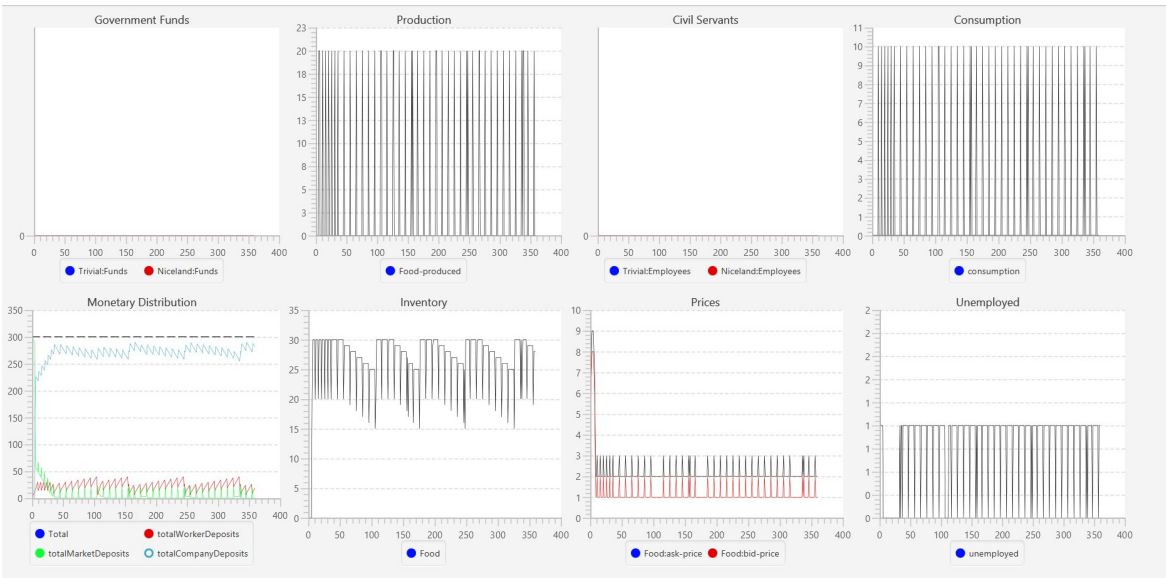

Figure 2.4: Trivial Economy with max spread 3

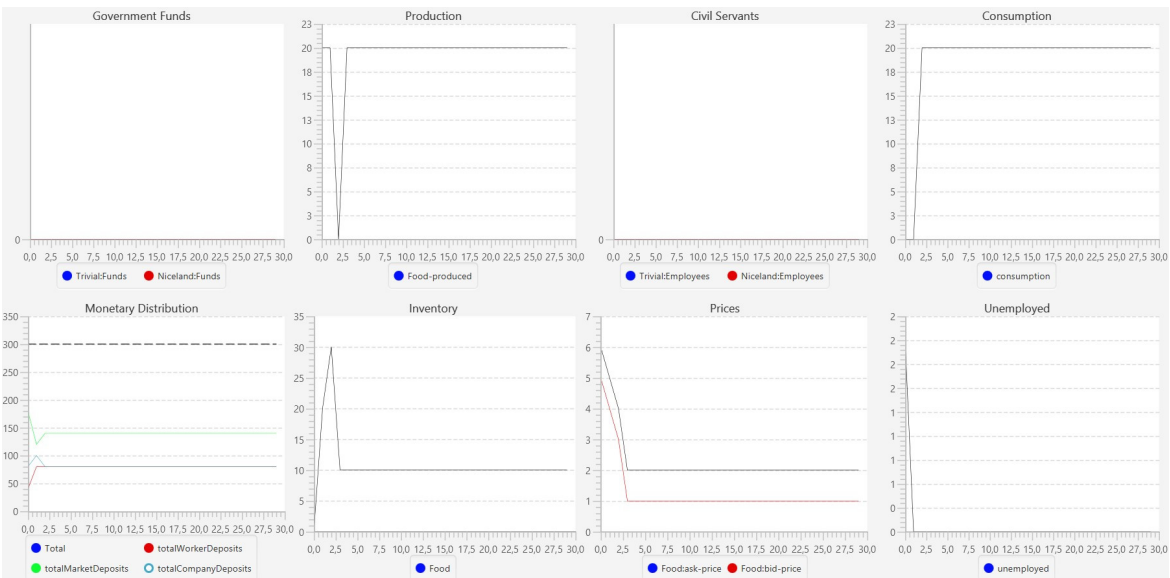

Figure 2.5: Trivial Economy with max spread 3

Agent	Input	Output
Government	$G_i = E \times Sal_{avg} \times T_p + (M_i + F_i) \times T_c$	$G_o = Sal_G \times E_G$
Market	$M_i = Q_c \times E \times P_{ask}$	$M_o = Sal_M \times E_M + M_i \times T_c$
Farm	$F_i = Q_c \times E \times P_{bid}$	$F_o = Sal_F \times E_F + F_i \times T_c$
Employee	$E_i = Sal$	$E_o = Q_c \times P_{ask} + Sal \times T_p$

Table 2.5: Village I/O

since there is still economic activity that continues in periodic cycles, but there is no longer constant consumption which means that the market is unable to supply the total demand. The reason it fails is because both the farm and the market end up paying their employees a salary of 2 per timestep. If the minimum salary is set to 20, which would be the minimum required since each person will have to buy 10 food and the lowest prices that the market can decide are bid prices of 1 and ask prices of 2, then the simulation finds the same price equilibrium, shown in figure 2.5.

2.3.2 Simulation: Small Village

The trivial economy was helpful in understanding how the market works and which parameters are required. Creating an economic simulation is an exercise in balancing the input and output flows of the agents while still trying to stay true to some kind of realism.

Now the simulation is expanded to have more people and a government to employ the people that do not work for the farm or the market, funded by income taxation. This simulation has 12 people that must consume 12 units of food per each time step; the number 12 is chosen due to having many factors which will make the division of labour simpler.

The labour output variable of the farm will tell us exactly how many workers are required to fulfill the total demand of 144. Assuming that the market will always sell at a higher price than the purchase price, and since the minimum bid:ask prices will be 1:2 each person agent must receive at least 24 income, after tax, to be able to keep consuming.

To decide which values we should assign to our variables, we can examine the input and output streams of each agent in table 2.5. Let's assume for the moment that there exists a market solution where every person is employed and is able to buy food to survive. That would mean that each input and output is, if not equal, at least close enough to oscillate around zero difference in the long run. To simplify, we can start with only levying personal income tax and set $T_c = 0\%$. Then the minimum salary must be

$$Sal_{min} \geq \frac{P_{ask} \times Q_c}{1 - T_p}$$

in order to suffice to buy food with income after tax; any value lower than that will never provide a solution.

The most important variable that we need to set is Lab, the labour output, because it will determine how large a part of the population is required to produce food to satisfy the total demand, which is $Q_c \times E$, where $E = E_G + E_M + E_F$ is the total number of employees (because everyone is employed). If we tacitly assume that each of the employers in the simulation, Government, Farm and Market, should be able to split the workforce evenly enough among themselves, we can set the labour output

Government		Market		Farm		Person x12	
Dep	1000	Dep	1000	Dep	1000	Dep	100
T_p	33	Emp	12	Lab	20	Sal	36
T_c	0	Inv	216			Q_p	12
Sal_{min}	36	Spr	1:5			Q_c	12
Civ	100	Pri	4:5			Q_s	12
Dem	0						

Table 2.6: Simulation: Small Village

Government		Market		Farm		Person x12	
Dep	1000	Dep	1000	Dep	1000	Dep	100
T_p	20	Emp	6	Lab	20	Sal	36
T_c	10	Inv	216			Q_p	12
Sal_{min}	36	Spr	1:5			Q_c	12
Civ	50	Pri	4:5			Q_s	12
Dem	0						

Table 2.7: Simulation: Small Village v2

to be $Lab = 3 \times Q_c$, then $E/3$ employees can produce the total demand. We don't have to put any restriction on the maximum allowed employees for the market and government and allow them to hire up to 100% of the workforce.

However, if the government agent has only personal income tax as revenue and we assume that the government salaries are close to the average $Sal_G \approx Sal_{avg}$ and we already know that $G_i \approx G_o$ is required in the constant money economy, then

$$G_i \approx G_o \implies E \times Sal_{avg} \times T_p \approx Sal_G \times E_G \implies E \times T_p \approx E \times Civ \implies T_p \approx Civ$$

which means there is a direct correlation between the personal income tax rate, T_p, and the percentage of civil servants, Civ. Even more interesting is the fact that if the solution is perfectly stable in each time step then

$$Sal_{avg} \times T_p = Sal_G \times Civ$$

and so, whenever $Sal_{avg} > Sal_G$, then $Civ > T_p$ must hold, and vice-versa.[2] From this we decide that a personal income tax ratio of 33% might be appropriate and calculate the minimum salary as 36.

Satisfied with our estimation of variables that could be likely to produce a successful simulation we try running it 10 times.

The longest simulation runs for 366 steps, the shortest only 15 steps and on average 110 steps, all ending when someone is unable to pay their personal income tax. In figure 2.6 we can see a typical result of these simulations. Solving the "order of evaluation" issues, as described by Mallett[7], is out of scope for this paper so a configuration that minimizes those issues was chosen. After some trial and error we find more stability by shifting some of the tax burden to the employers and put a maximum 50% on the number of employees for the market and the government.

Most of the simulations still end after a short time with the same error but we also do get some runs like in figure 2.7, which shows the status after 485 steps where it

[2]This is not highly relevant to our simulation but we will leave it to economists and politicians to debate the implications of this.

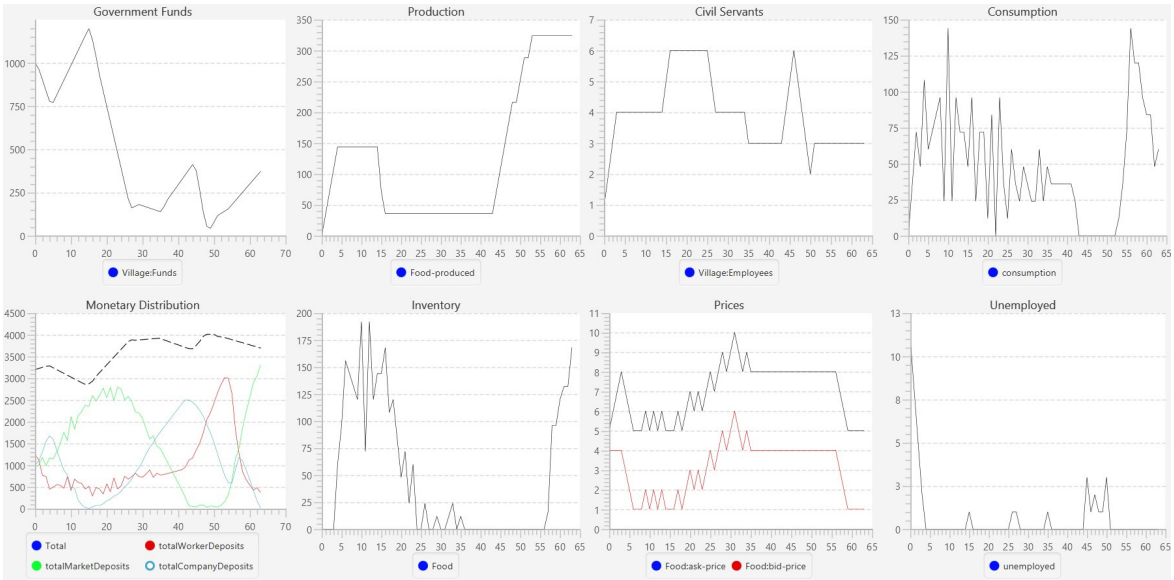

Figure 2.6: Small Village

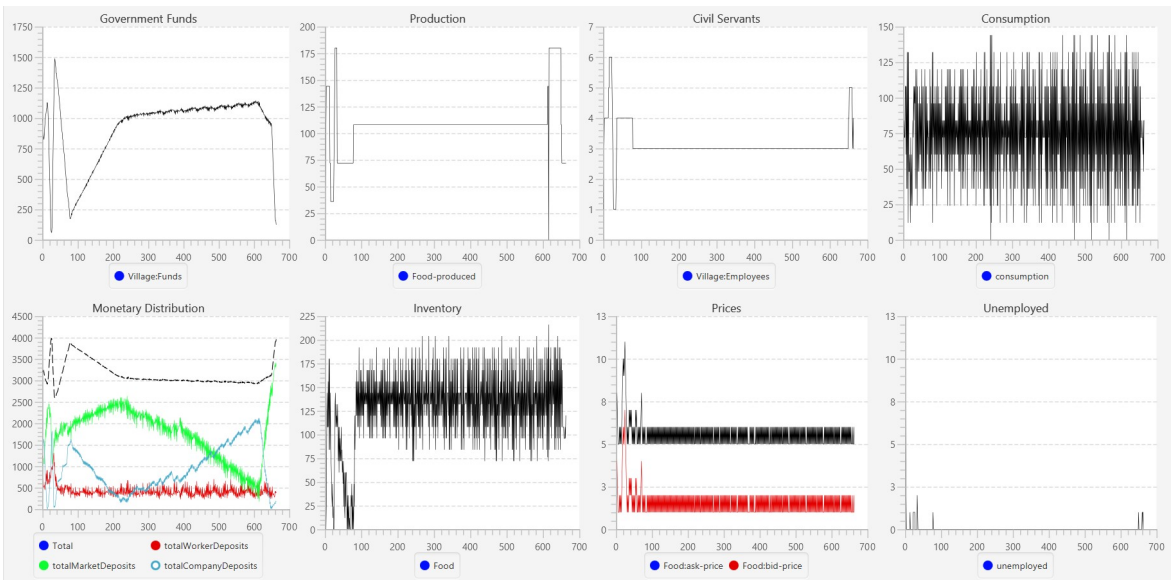

Figure 2.7: Small Village v2

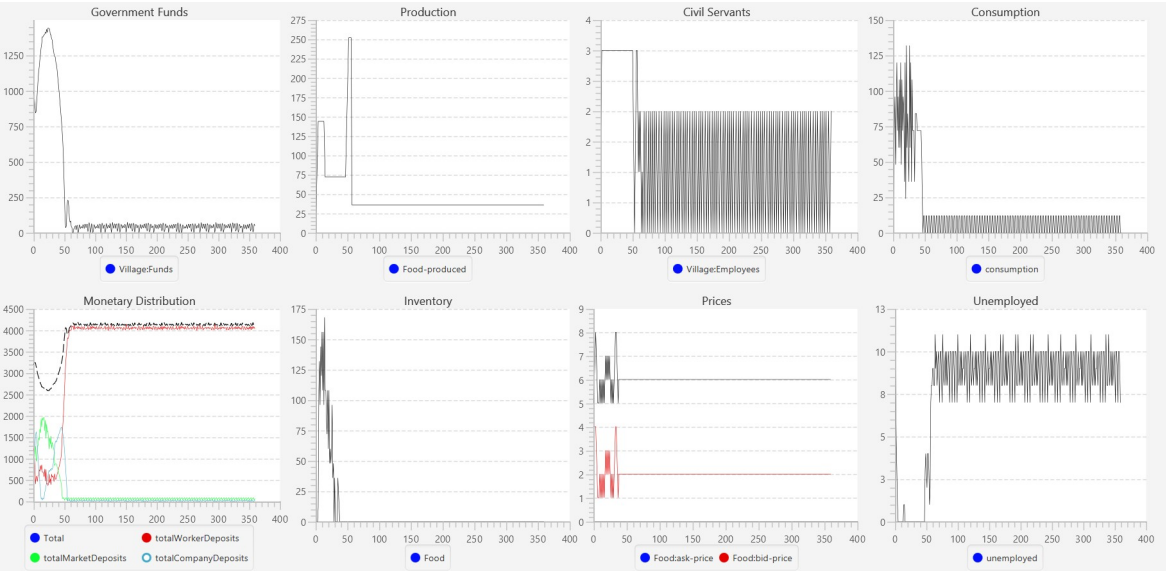

Figure 2.8: Small Village v3

still fails on personal tax payments. We notice a relatively stable period with 3 civil workers so we try to enforce that, setting $Civ = 25\%$ (and reset the market employees as $Emp = 12$) but then most simulations end with the money accumulating with the employees, the market runs out of money and is unable to buy food from the market so there is nowhere for the money to go, see figure 2.8.

Chapter 3

Results

Now we will implement demurrage into our previous simulations and analyze the effects it has on the simulated economy.

3.1 Small Village: Demurrage 1%

We start by setting a 1% nominal demurrage step $Dem = 1$, so that in each time step, 1% of all personal and company deposits are taxed, and run the simulation 10 times. Most of them are still prematurely ended due to insufficient funds to pay taxes, the shortest are 11 steps, average 458 steps but the longest we see is 2637 steps, which would be over 7.25 years if each step was one day. See figure 3.1.

We can see some cyclic behavior here, with increased demurrage income the government is able to hire more employees that buy food from the market, which keeps the production going, although it is not constant.

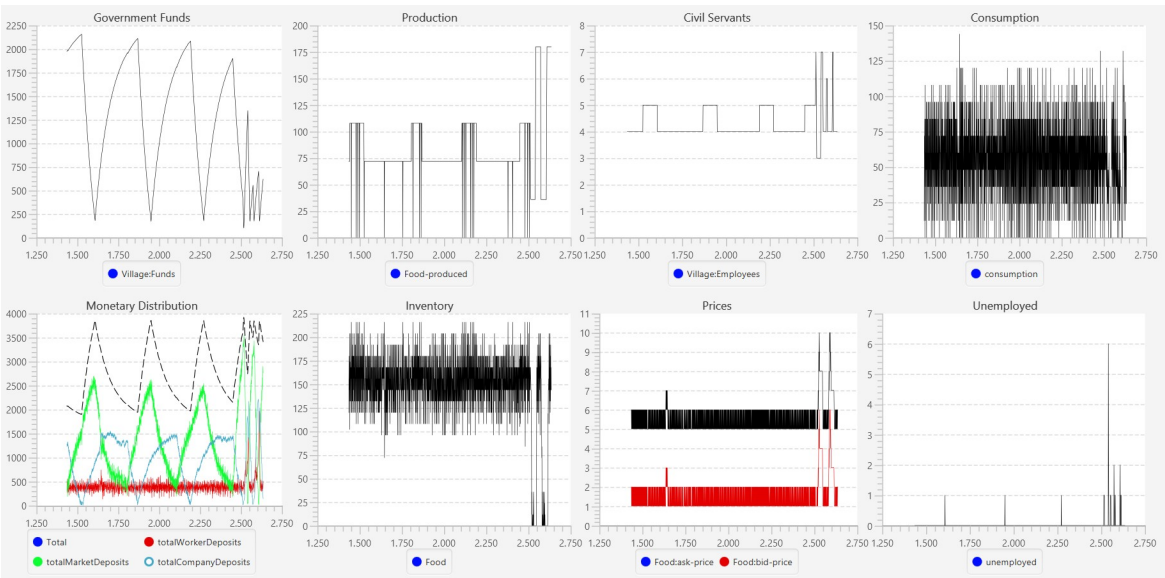

Figure 3.1: Demurrage 1%

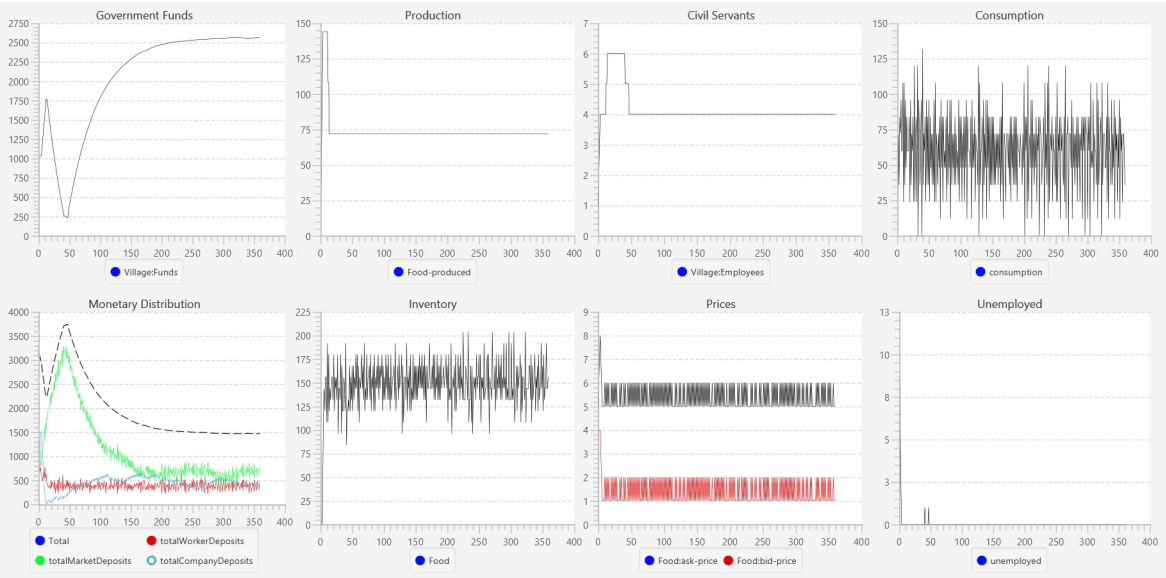

Figure 3.2: Demurrage 2%

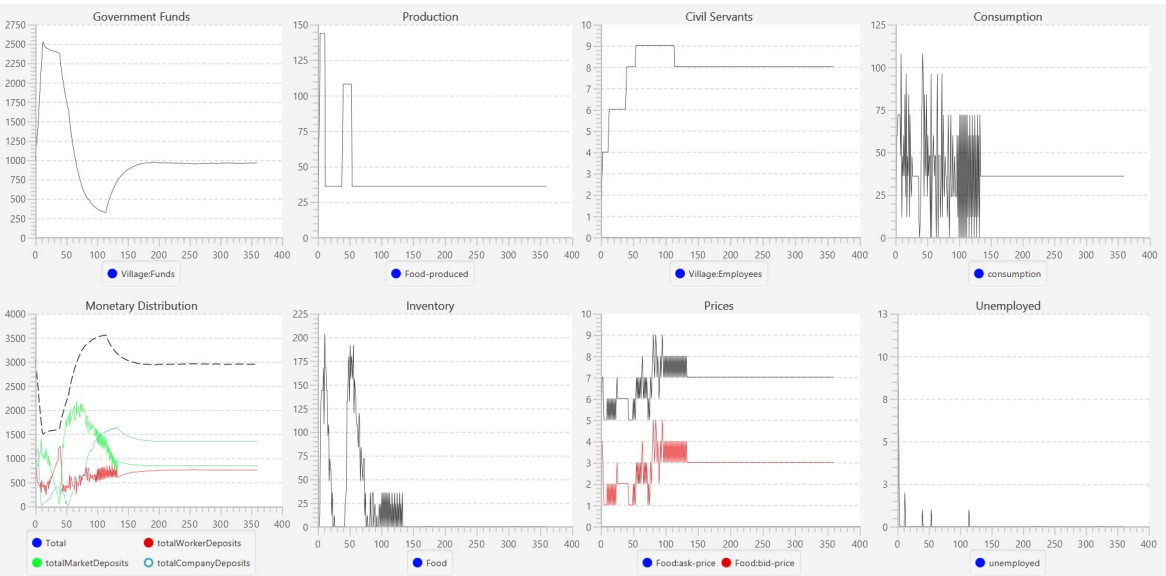

Figure 3.3: Demurrage 5%

3.2 Small Village: Demurrage 2%

With 2% demurrage we start to see a lot more stability. The amounts per agent may wary but they tend to line out at some point, as seen in figure 3.2.

3.3 Small Village: Demurrage 5%

Increasing the demurrage to 5% most simulations tend to stabilize, either with a flat-line as in figure 3.3, a cyclic balance as shown in figure 3.4, but also in a failure where the employees hoard the money as in figure 3.5. The higher demurrage seems to bring more stability in the simulation but it is questionable whether this could be a realistic amount in the real economy.

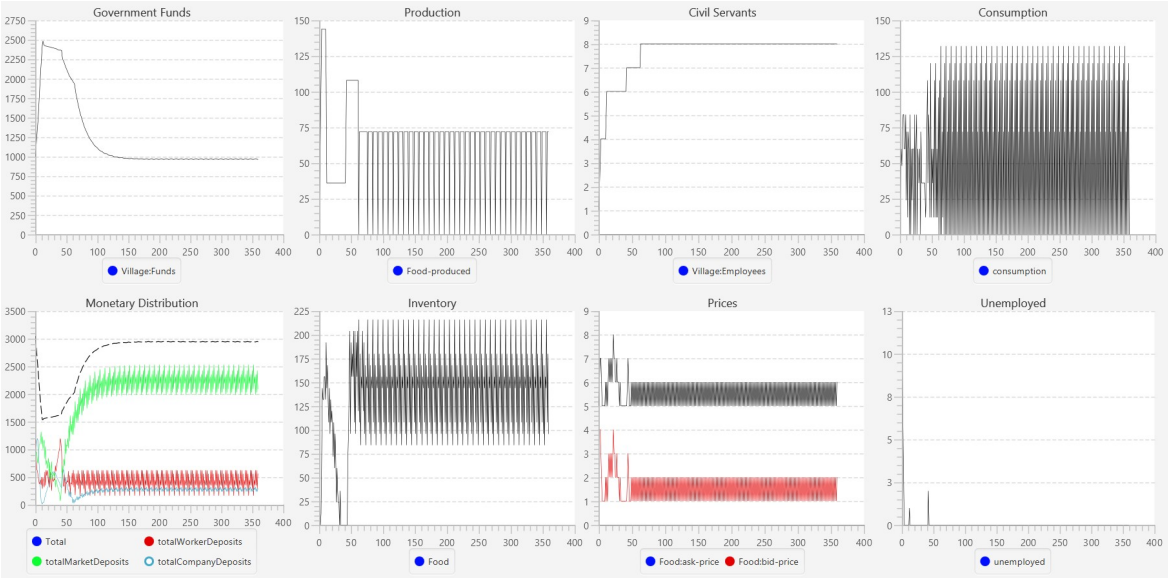

Figure 3.4: Demurrage 5%

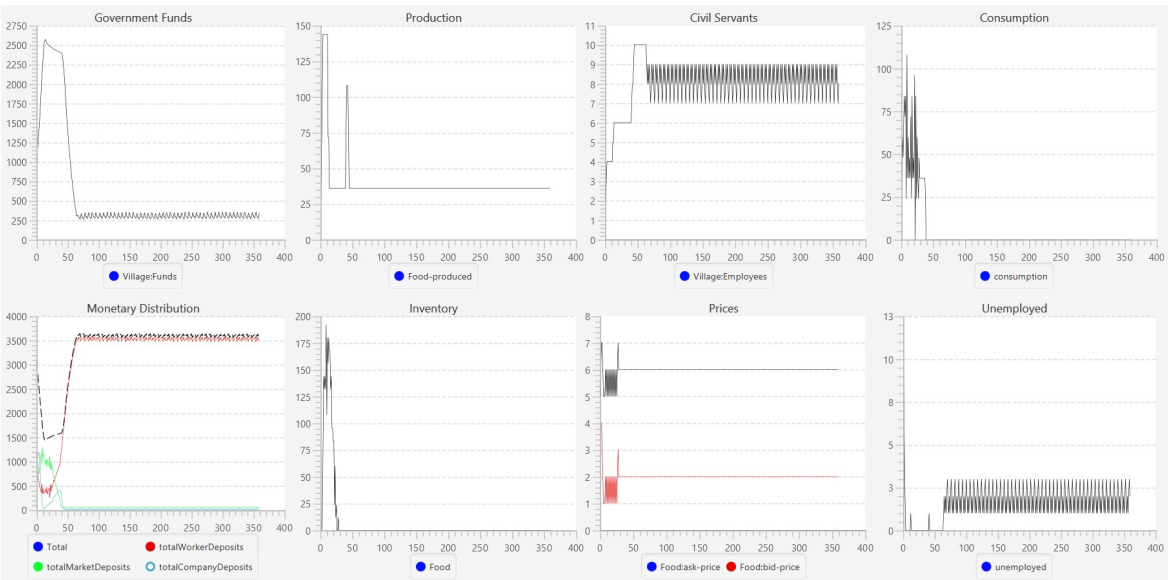

Figure 3.5: Demurrage 5%

Chapter 4

Discussion

4.1 Summary

We have successfully implemented currency demurrage into Threadneedle as a tax on the balance of deposit accounts. Demurrage adds a new output stream of money to the agents in the simulation, decreasing the chances of money accumulating and ending the simulation. It provides the government with a new income stream that allows for increased expenditure for civil servants.

Designing an economic simulation is not trivial, even in an extremely simplified environment like we have constructed in Threadneedle. Many of the variables that affect the simulation have to be decided, approximated from real world values, abstracted away or just arbitrarily assigned. Defining a success metric of the simulated economies also proved challenging. It is conceivable that a perfect simulation of a real world economy actually does not work according to any definition we would like to apply to our simulation. The obvious breaking point of the simulation is when an agent must spend money they do not have. As we have mentioned, every agent must be able to satisfy their output stream with an input for the simulation to continue. But what should happen in cases where this is not satisfied. If we think about this in real world terms then a company that becomes insolvent might declare bankruptcy and thereby effectively cease to exist. We currently don't have an implementation for that in Threadneedle, which might be a good future addition, so we will have to view that as a failure in the simulation. Similarly, the workers might be unable to purchase the food they require for consumption if they don't have the income. In the real world a big part of our economy and society in general revolves around trying to make sure that people will not starve to death even if they aren't able to afford food, whether that is accomplished with charities, social service or money lending.

4.2 Conclusion

Adding demurrage to an existing economy adds a new revenue stream for the government agent and prevents hoarding of money by other agents in the system. In simple simulations, as we have shown here, where the avenues for money to flow through are limited, this has a positive effect on the simulation since preventing hoarding is likely to prolong the simulation. We don't think it is prudent to draw any conclusions about implementing demurrage in the real world from these simulations, due to their limited scale, but our results seem to have a positive effect within the parameters we set up.

Positive here means extending the duration of the simulation before agents are unable to pay for needed expenses given their revenue.

Building the baseline economies proved more difficult than expected, since minor changes to initial variables often resulted in very different and sometimes unexpected results.

4.3 Future work

More work needs to be done to create larger and more complex simulations using demurrage. Creating a baseline economy that would more closely resemble a real-world economy might also give better insight into how demurrage would behave in the real world. The simulations could have multiple types of products and producers, more than one bank and even multiple governments over several regions.

We have currently only looked at demurrage with a constant money supply in our simulation but an obvious next step would be to implement demurrage in a fractional reserve economy, where the total money supply can expand (and contract). Effects of demurrage on the liquidity issues of banks would have to be examined as this would affect their reserves. Of special interest would be to measure if adding demurrage would affect inflation in an economy, if it would increase or decrease and by how much.

Using demurrage to fund a universal basic income in Threadneedle would also force a certain circulation of money through the system which could yield interesting results. It would most likely allow the simulations to run longer since most of the failures arise when the circulation stops.

Different implementations of demurrage could be explored, e.g. the table money where the value of the physical currency decreases over time effectively reducing the money supply. If money is printed at the same rate as the table money increases then a direct comparison with the taxation implementation could show whether or not they are equivalent. Or perhaps even if they are mathematically equivalent, they might not be politically equivalent depending on how intuitive the mechanisms are to the public.

Threadneedle provides an interesting avenue of economic research with the potential to examine if multi-agent systems, where agents continually make micro-adaptations based on their discrete knowledge of the whole system, can be expected to produce stability and then which variables are correlated with that stability.

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Appendix A

Code

Listing A.1: Govt.java: Adding demurrage function.

```
public boolean payDemurrage(Account fromAccount, long deposit)
{
  long amountToPay = (long) (deposit * demurrageRate / 100.0);

  if (amountToPay > 0) {
    //System.out.println("Paying demurrage: " + fromAccount.getName() + "
      - " + amountToPay);
  }

  s_totalRevenue.add(amountToPay);
  return payGovtTax(fromAccount, amountToPay);
}
```

Listing A.2: trivial.json


```
{
"description" :"",
"GsonAgent" :
 [
  {"clss":"BaselGovt","json":"{\"country\":\"Trivial\",\"capitalPct\":0.0,\"reservePct\":0,\"reserveControls\":false,\"capitalControls\":false,\"payUnemployment\":false,\"unemploymentPayment\":0,\"payUBI\":false,\"ubiPayment\":0,\"personalTaxRate\":0,\"corporateTaxRate\":0,\"personalCutoff\":0,\"corporateCutoff\":0,\"maxCivilServants\":0,\"treasuryRate\":0.0,\"minWage\":0,\"demurrageRate\":0,\"name\":\"Niceland\",\"initialDeposit\":0,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
 {"clss":"Bank","json":"{\"sharePrice\":10,\"capitalPct\":0.0,\"capitalDividend\":0.0,\"capitalSteps\":12,\"interestRateDelta\":0,\"writeOffLimit\":6,\"lossProvisionPct\":0.0,\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Bank-4\",\"initialDeposit\":0,\"bankname\":\"\",\"x\":380.0,\"y\":440.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"LabourMarket","json":"{\"maxEmployees\":1,\"bidEqualsAsk\":false,\"maxInventory\":20,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":2,\"payDividend\":false,\"product\":\"Labour\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Labour\",\"initialDeposit\":0,\"bankname\":\"Central Bank\",\"x\":400.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Market","json":"{\"maxEmployees\":1,\"bidEqualsAsk\":false,\"maxInventory\":3,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":2,\"payDividend\":false,\"product\":\"Food\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"M-Food\",\"initialDeposit\":30,\"bankname\":\"Bank-4\",\"x\":400.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Farm","json":"{\"product\":\"Food\",\"labourInput\":2,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Farm-5\",\"initialDeposit\":0,\"bankname\":\"Bank-4\",\"x\":446.0,\"y\":193.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Person","json":"{\"profile\":{\"needs\":{\"Food\":{\"product\":\"Food\",\"quantity\":1,\"storeQ\":1,\"consumption\":1,\"stepFrequency\":0,\"useLoan\":false,\"consumable\":true}}},\"desiredSalary\":2,\"randomPurchase\":false,\"name\":\"ID-8\",\"initialDeposit\":0,\"bankname\":\"Bank-4\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
```

Listing A.3: village.json

```
{
"description" :"",
"GsonAgent" :
 [
  {"clss":"BaselGovt","json":"{\"country\":\"Village\",\"capitalPct\":0.0,\"reservePct\":0,\"reserveControls\":false,\"capitalControls\":false,\"payUnemployment\":false,\"unemploymentPayment\":0,\"payUBI\":false,\"ubiPayment\":0,\"personalTaxRate\":33,\"corporateTaxRate\":0,\"personalCutoff\":0,\"corporateCutoff\":0,\"maxCivilServants\":100,\"treasuryRate\":0.0,\"minWage\":36,\"name\":\"Village\",\"initialDeposit\":1000,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
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  {"clss":"LabourMarket","json":"{\"maxEmployees\":1,\"bidEqualsAsk\":false,\"maxInventory\":20,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Labour\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Labour\",\"initialDeposit\":0,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Market","json":"{\"maxEmployees\":12,\"bidEqualsAsk\":false,\"maxInventory\":216,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Food\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"M-Food\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":420.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Farm","json":"{\"product\":\"Food\",\"labourInput\":36,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Farm-5\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":456.0,\"y\":172.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Person","json":"{\"profile\":{\"needs\":{\"Food\":{\"product\":\"Food\",\"quantity\":12,\"storeQ\":12,\"consumption\":12,\"stepFrequency\":0,\"useLoan\":false,\"consumable\":true}}},\"desiredSalary\":36,\"randomPurchase\":false,\"name\":\"ID-7\",\"initialDeposit\":100,\"bankname\":\"Bank-4\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
```

Listing A.4: villagev2.json

```
{
"description" :"",
"GsonAgent" :
 [
  {"clss":"BaselGovt","json":"{\"country\":\"Village\",\"capitalPct\":0.0,\"reservePct\":0,\"reserveControls\":false,\"capitalControls\":false,\"payUnemployment\":false,\"unemploymentPayment\":0,\"payUBI\":false,\"ubiPayment\":0,\"personalTaxRate\":20,\"corporateTaxRate\":10,\"personalCutoff\":0,\"corporateCutoff\":0,\"maxCivilServants\":50,\"treasuryRate\":0.0,\"minWage\":36,\"name\":\"Village\",\"initialDeposit\":1000,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
 {"clss":"Bank","json":"{\"sharePrice\":0,\"capitalPct\":0.0,\"capitalDividend\":0.0,\"capitalSteps\":12,\"interestRateDelta\":0,\"writeOffLimit\":6,\"lossProvisionPct\":0.0,\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Bank-4\",\"initialDeposit\":0,\"bankname\":\"\",\"x\":298.0,\"y\":219.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"LabourMarket","json":"{\"maxEmployees\":1,\"bidEqualsAsk\":false,\"maxInventory\":20,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Labour\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Labour\",\"initialDeposit\":0,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Market","json":"{\"maxEmployees\":6,\"bidEqualsAsk\":false,\"maxInventory\":216,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Food\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"M-Food\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":420.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Farm","json":"{\"product\":\"Food\",\"labourInput\":36,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Farm-5\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":456.0,\"y\":172.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Person","json":"{\"profile\":{\"needs\":{\"Food\":{\"product\":\"Food\",\"quantity\":12,\"storeQ\":12,\"consumption\":12,\"stepFrequency\":0,\"useLoan\":false,\"consumable\":true}}},\"desiredSalary\":36,\"randomPurchase\":false,\"name\":\"ID-7\",\"initialDeposit\":100,\"bankname\":\"Bank-4\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
```

Listing A.5: villagev3.json

```
{
"description" :"",
"GsonAgent" :
 [
  {"clss":"BaselGovt","json":"{\"country\":\"Village\",\"capitalPct\":0.0,\"reservePct\":0,\"reserveControls\":false,\"capitalControls\":false,\"payUnemployment\":false,\"unemploymentPayment\":0,\"payUBI\":false,\"ubiPayment\":0,\"personalTaxRate\":20,\"corporateTaxRate\":10,\"personalCutoff\":0,\"corporateCutoff\":0,\"maxCivilServants\":25,\"treasuryRate\":0.0,\"minWage\":36,\"name\":\"Village\",\"initialDeposit\":1000,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
 {"clss":"Bank","json":"{\"sharePrice\":0,\"capitalPct\":0.0,\"capitalDividend\":0.0,\"capitalSteps\":12,\"interestRateDelta\":0,\"writeOffLimit\":6,\"lossProvisionPct\":0.0,\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Bank-4\",\"initialDeposit\":0,\"bankname\":\"\",\"x\":298.0,\"y\":219.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"LabourMarket","json":"{\"maxEmployees\":1,\"bidEqualsAsk\":false,\"maxInventory\":20,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Labour\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Labour\",\"initialDeposit\":0,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Market","json":"{\"maxEmployees\":12,\"bidEqualsAsk\":false,\"maxInventory\":216,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Food\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"M-Food\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":420.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Farm","json":"{\"product\":\"Food\",\"labourInput\":36,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Farm-5\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":456.0,\"y\":172.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Person","json":"{\"profile\":{\"needs\":{\"Food\":{\"product\":\"Food\",\"quantity\":12,\"storeQ\":12,\"consumption\":12,\"stepFrequency\":0,\"useLoan\":false,\"consumable\":true}}},\"desiredSalary\":36,\"randomPurchase\":false,\"name\":\"ID-7\",\"initialDeposit\":100,\"bankname\":\"Bank-4\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
```


Listing A.6: villagedem1.json


```
{
"description" :"",
"GsonAgent" :
 [
  {"clss":"BaselGovt","json":"{\"country\":\"Village\",\"capitalPct\":0.0,\"reservePct\":0,\"reserveControls\":false,\"capitalControls\":false,\"payUnemployment\":false,\"unemploymentPayment\":0,\"payUBI\":false,\"ubiPayment\":0,\"personalTaxRate\":33,\"corporateTaxRate\":0,\"personalCutoff\":0,\"corporateCutoff\":0,\"maxCivilServants\":100,\"treasuryRate\":0.0,\"minWage\":36,\"demurrageRate\":1,\"name\":\"Village\",\"initialDeposit\":1000,\"bankname\":\"Central Bank\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
 {"clss":"Bank","json":"{\"sharePrice\":0,\"capitalPct\":0.0,\"capitalDividend\":0.0,\"capitalSteps\":12,\"interestRateDelta\":0,\"writeOffLimit\":6,\"lossProvisionPct\":0.0,\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Bank-4\",\"initialDeposit\":0,\"bankname\":\"\",\"x\":400.0,\"y\":440.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"LabourMarket","json":"{\"maxEmployees\":1,\"bidEqualsAsk\":false,\"maxInventory\":20,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Labour\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":50,\"name\":\"Labour\",\"initialDeposit\":0,\"bankname\":\"Central Bank\",\"x\":400.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Market","json":"{\"maxEmployees\":12,\"bidEqualsAsk\":false,\"maxInventory\":216,\"ttl\":-1,\"minSpread\":1,\"maxSpread\":5,\"payDividend\":false,\"product\":\"Food\",\"labourInput\":0,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"M-Food\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":400.0,\"y\":400.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Farm","json":"{\"product\":\"Food\",\"labourInput\":36,\"salaryPeriod\":1,\"period\":1,\"minCapital\":0,\"name\":\"Farm-5\",\"initialDeposit\":1000,\"bankname\":\"Bank-4\",\"x\":456.0,\"y\":172.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
  {"clss":"Person","json":"{\"profile\":{\"needs\":{\"Food\":{\"product\":\"Food\",\"quantity\":12,\"storeQ\":12,\"consumption\":12,\"stepFrequency\":0,\"useLoan\":false,\"consumable\":true}}},\"desiredSalary\":36,\"randomPurchase\":false,\"name\":\"ID-7\",\"initialDeposit\":100,\"bankname\":\"Bank-4\",\"x\":0.0,\"y\":0.0,\"defaultProb\":0.0,\"regionName\":\"\"}"},
```